Journal of Maps

ISSN: 1744-5647 (Online) Journal homepage: www.tandfonline.com/journals/tjom20

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To cite this article: Konstantinos Tsanakas, Efthimios Karymbalis, Diamantina Griva, Kanella Valkanou, Dimitrios-Vasileios Batzakis, Emmanuel Vassilakis, Aliki Konsolaki & Kalliopi Gaki-Papanastassiou (2025) The Geomorphology of Greece, Journal of Maps, 21:1, 2540555, DOI: 10.1080/17445647.2025.2540555

To link to this article: https://doi.org/10.1080/17445647.2025.2540555

JOURNAL OF MAPS
2025, VOL. 21, NO. 1, 2540555
https://doi.org/10.1080/17445647.2025.2540555

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The Geomorphology of Greece

Konstantinos Tsanakas[a*], Efthimios Karymbalis[a*], Diamantina Griva[a], Kanella Valkanou[a], Dimitrios-Vasileios Batzakis[a], Emmanuel Vassilakis[b], Aliki Konsolaki[b] and Kalliopi Gaki-Papanastassiou[b]

[a]Department of Geography, Harokopio University, Athens, Greece; [b]Department of Geography and Climatology, Faculty of Geology and Geoenvironment, National and Kapodistrian University of Athens, Athens, Greece

ABSTRACT

The study presents a geomorphological overview of Greece at a 1:1,000,000 scale, marking the first national cartographic synthesis effort to interpret geological and geomorphological factors shaping the country's landscape. The geomorphological map was developed through a literature review of prior studies at varying scales and semi-automated GIS techniques. High-resolution topographic data and 1:50,000 geological maps were integrated into a spatial geodatabase. Secondary layers, including a hillshade map, slope-aspect map, and red relief image map, were created and combined with Google Earth imagery to delineate landforms. These were categorized by primary formative processes into structural, fluvial, gravity-induced, coastal, karst, volcanic, glacial and periglacial. Additional maps and tables were produced, detailing topographic parameters, geotectonic setting, and climatic regime. The results highlight Greece's diverse geomorphological environments, shaped by active tectonics and surface processes. The map represents recent geomorphological advancements and serves as a management tool for stakeholders and a reference for interdisciplinary research.

ARTICLE HISTORY
Received 2 January 2025
Revised 5 June 2025
Accepted 23 July 2025

KEYWORDS
GIS; landforms; landscape classification; national scale; geomorphological map

1. Introduction

Geomorphology plays a pivotal role in Earth-related disciplines, providing insights into landscape formation and evolution (Sherman, 1989). Geomorphological mapping is a significant tool that enhances our understanding of natural processes shaping the Earth's landscape over time and disseminates this knowledge across scientific fields. In recent decades, it finds extensive application in various sectors of environmental research, including disaster risk assessment, natural resource management, and ecosystem conservation. Additionally, geomorphological maps support essential tasks in civil engineering, urban planning projects, transportation infrastructure development, and land-use zoning (Griffiths & Marsh, 1986). They serve as key elements for decision-making in sustainable urban development, helping in the planning of resilient cities and reduction of environmental impacts. This wide-ranging interdisciplinary application reflects the growing demand for the crucial data they provide (Gustavsson et al., 2006). As a result, geomorphological mapping has become a standard and widely used preliminary investigation method in disciplines related to land management, environmental protection, and sustainable development (Dramis et al., 2011).

In recent years there has been a significant increase in the production of large-scale geomorphological maps, reflecting their essential role in advancing geomorphological research and related fields (Adeli & Khorshiddoust, 2011; Arnaud-Fassetta, 2003; Batzakis et al., 2020; Kalogeropoulos et al., 2023; Karymbalis et al., 2021, 2022a, 2022b, 2022c, 2022d; Luino et al., 2018; Tsanakas et al., 2016; Valkanou et al., 2022). Although there is a relatively large number of large-scale maps, small-scale maps are comparatively fewer, as geomorphological maps at smaller scales are more complex to conduct. Even though many nations have accomplished their geological mapping programs at a national scale, only a limited number of them have established nationwide geomorphological mapping initiatives (Smith et al., 2011). Noteworthy examples encompass Spain, Italy, Romania, and Poland (Buza, 1997; Rączkowska & Zwoliński, 2015; Salazar Rincón & Martín-Serrano García, 2006; Servizio Geologico, Nazionale, n.d.). Russia has implemented a comprehensive geomorphological mapping program, while Australia's recent regolith mapping program relies significantly on geomorphological mapping principles (Regolith Landscape Evolution across Australia: Monographs: Publications:

CONTACT Konstantinos Tsanakas ktsanakas@hua.gr Department of Geography, Harokopio University, 70 El. Venizelou Str., Athens GR-17676, Greece
*Shared first authorship.

Supplemental map for this article can be accessed online at https://doi.org/10.1080/17445647.2025.2540555.

CRC LEME, n.d.). Additionally, China has recently introduced a geomorphological atlas at a national scale (Cheng et al., 2011), while Brazil in South America, is notable for its extensive geomorphological mapping, focusing mainly on the Amazon rainforest, and the arid regions of the country (Ab'Saber, 1969).

In Greece, a relatively limited number of geomorphological maps have been published. Pavlopoulos (1988) produced a geomorphological map of southern Attica at a scale of 1:25.000 aiming to identify the morphogenetic processes that shaped the landscape during the Quaternary. Karymbalis (1996) compiled a detailed geomorphological map of the Evinos River delta to document the main morphological features and investigate the natural processes that shape the deltaic plain of the River. Karymbalis et al. (2013) produced the geomorphological map of Cephalonia Isl. in Western Greece at a scale of 1:50,000, utilizing field surveys and GIS analysis. Subsequently, Karymbalis et al. (2016) presented a geomorphological map of the Pinios River delta in central Greece at a scale of 1:15,000, reflecting the results of extensive fieldwork combined with GIS and aerial photos analyses. To understand and interpret the geomorphological factors driving the formation and evolution of the Pieria Mtns in Northern Greece, Tsanakas et al. (2019a, 2019b) employed both GIS-oriented and traditional mapping techniques to produce a 1:50,000 scale geomorphological map of the region. Finally, Chabrol et al. (2022) carried out detailed geomorphological mapping of the Kalamas River delta by combining field surveys with the interpretation of historical maps satellite imagery and aerial photographs. These studies demonstrate the advancement of geomorphological mapping in Greece, emphasizing the variety of methods and scales applied to understand the processes shaping its landscapes.

Over the last decades, the geomorphology of Greece has been extensively studied. Despite the abundance of geomorphological information derived from research on topics such as tectonic geomorphology, fluvial and coastal geomorphology, Quaternary paleogeographic reconstruction etc. (Karymbalis et al., 2018a, 2018b; Karymbalis et al., 2022; Tsanakas et al., 2019a, 2019b; Tsanakas et al., 2022a, 2022b), there has been a notable lack of efforts to integrate this data into a comprehensive, national-scale geomorphological map. The challenge lies in the high level of generalization required to depict landforms originally mapped at larger scales, hindering the integration of this extensive geomorphological knowledge into a cohesive, national-scale map. However, recognizing the importance of small-scale geomorphological mapping for understanding local-scale processes and landform dynamics, this study endeavors to address these challenges. In this context, this study presents a geomorphological overview of Greece at a scale of 1:1,000,000, marking an innovative attempt to compile and interpret geological and geomorphological processes that have shaped the Greek landscape over geological time. The resulting Main Map serves as a valuable tool for evaluating landscape evolution and provides a robust foundation for further interdisciplinary analysis.

2. Study area

2.1 Location and general setting

This study focuses on the geomorphological mapping of Greece, located in southeast Europe between approximately 35.8° to 41.8° latitude and 19.4° to 28.2° longitude (Figure 1). More than 80% of the country exhibits a mountainous or hilly terrain. The highest elevation is found at Mt. Olympus, reaching 2,917 m. In addition to its mountainous terrain, Greece also features extensive plains, primarily located in the central and northern regions of the country. Its coastline, including a large number of islands, has a length of 13,676 km. With a rich geomorphological diversity, Greece exhibits a variety of landforms, including tectonic, fluvial, coastal, karst, structural, glacial and periglacial, and many others.

2.2 Geological and tectonic setting

Greece is a tectonically and seismically highly active region, positioned near convergent tectonic plate boundaries (Figure 2a,b). Its most significant tectonic structures that play a crucial role in the region's complex geodynamics include the Hellenic Subduction Zone, the North Anatolian Fault (NAF) system, the Hellenic Arc, the Pindos Orogenic Belt, the Cephalonia Transform Fault Zone (CTFZ), and the Corinth Rift.

The Hellenic Subduction Zone is the most prominent tectonic feature in the Eastern Mediterranean. It involves the subduction of the African Plate beneath the Eurasian Plate at a mean rate of ~36 mm a^{-1} (McClusky et al., 2000; Reilinger et al., 2006), forming the Hellenic Arc and Trench. Data from earthquake analyses, seismic tomography, and seismic reflection studies demonstrate that the Hellenic Subduction Zone is the most extensive, rapidly moving, and seismically active subduction zone in the Mediterranean region (Papazachos et al., 2000; Wortel & Spakman, 2000). GPS measurements indicate that the rates of crustal movement offshore of the Peloponnese and Crete Island, relative to Africa, can reach up to approximately 40 mm a^{-1} (Hollenstein et al., 2008; Kahle et al., 2000).

The North Anatolian Fault (NAF) System is a strike-slip right lateral fault which extends from Türkiye into the North Aegean Sea and terminates in

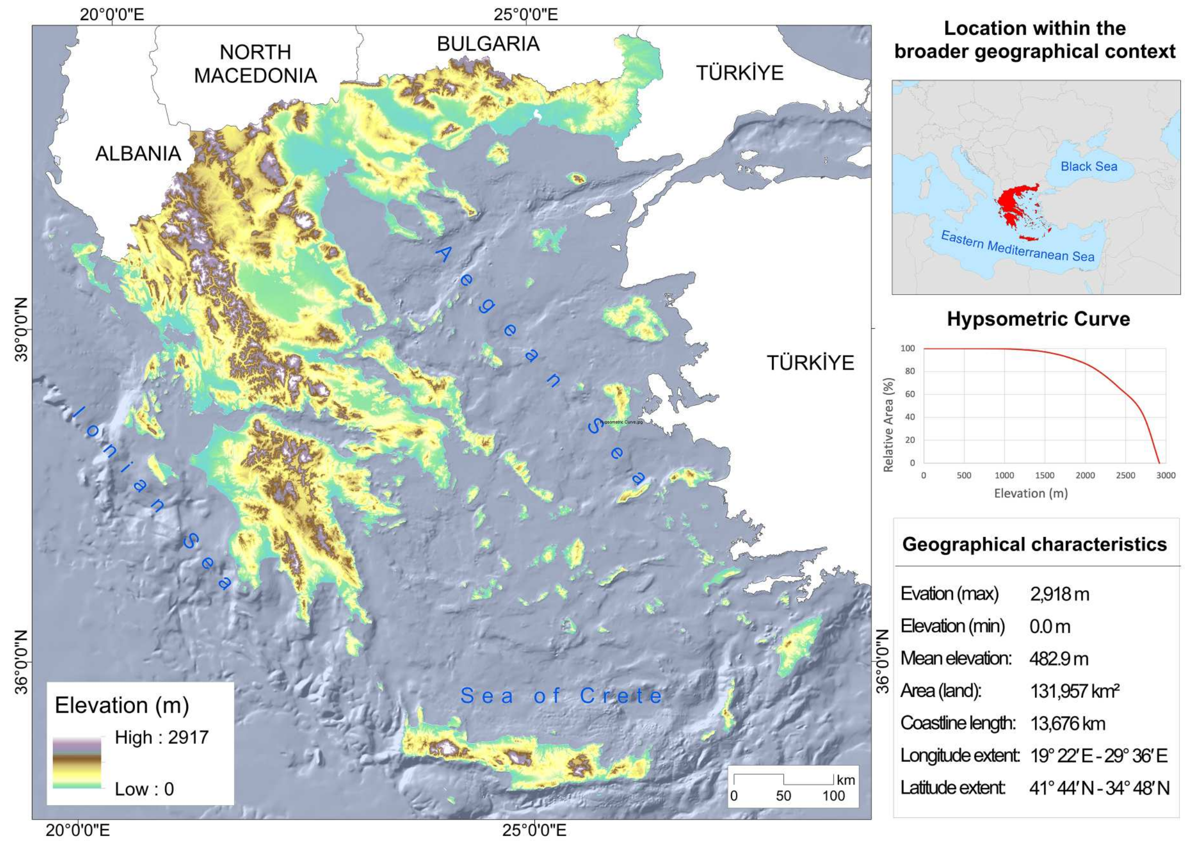


Figure 1. Digital Elevation Model (DEM) of Greece, illustrating topographic variation. An inset map depicts the location of the country within Europe. The figure includes the hypsometric curve, showing the distribution of elevations across the country, as well as key geographic metrics.

the North Aegean Trough (Ferentinos et al., 2018; Rolando et al., 1999).

The Hellenic Arc is the result of the Hellenic Subduction Zone. It is a prominent arc-shaped structure that includes both an outer sedimentary arc and an inner volcanic arc (Le Pichon et al., 2019; Papanikolaou, 2013; Ring et al., 2010).

The Pindos Orogenic Belt is a major mountain range in mainland Greece, representing an older tectonic structure formed during the Alpine orogeny (Botziolis et al., 2023). It consists of folded and faulted sedimentary rocks, primarily from the Mesozoic and Cenozoic eras (Papanikolaou, 2021b). Although tectonic activity has shifted towards extensional

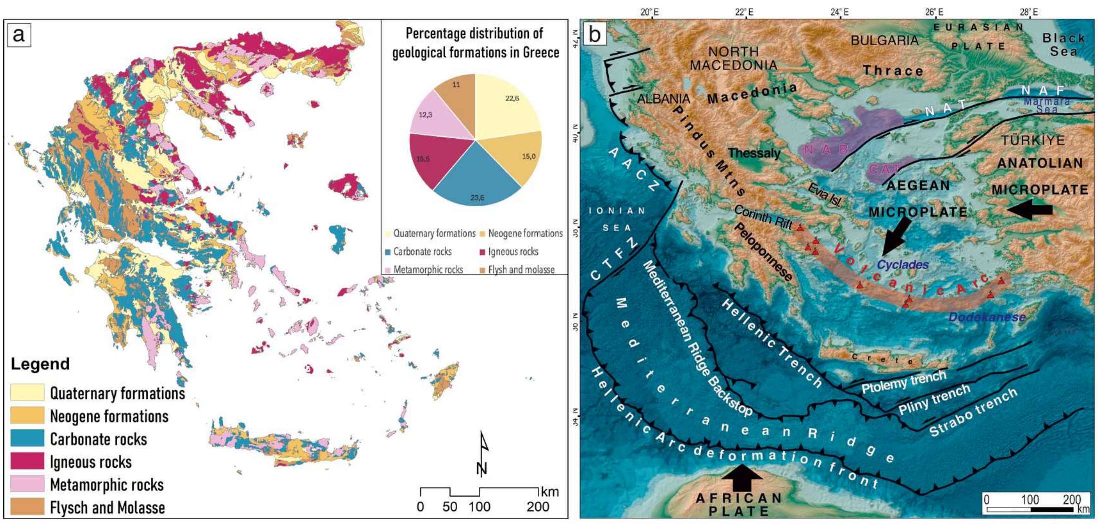


Figure 2. a) Distribution of geological formations in Greece (Hellenic Survey of Geology and Mineral Exploration) b) The main geotectonic structures of the broader Aegean region. AACZ: Adria-Aegean Collision Zone, CTFZ: Cephalonia Transform Fault Zone, CAT: Central Aegean Trough, NAB: North Aegean Basin, NAF: North Anatolian Fault, NAT: North Aegean Trough (modified from (Reilinger et al., 2006; Vött, 2007)).

processes, the Pindos range remains a significant structural feature (Roberts & Jackson, 1991).

The Cephalonia Transform Fault Zone (CTFZ), located in the Ionian Sea, west of Cephalonia Island, separates the Hellenic subduction zone to the south from the more tectonically stable Adriatic microplate to the north (Louvari et al., 1999). The CTFZ is associated with strong seismic events and influences the tectonic deformation in the western part of Greece (Sachpazi et al., 2000).

The Corinth Rift is one of the most active extensional tectonic regions in Europe, characterized by rapid crustal stretching and normal faulting (Armijo et al., 1996; Bell et al., 2009; Fernández-Blanco et al., 2019; Moretti et al., 2003; Rolando et al., 1999; Taylor et al., 2011). It is a rift zone formed due to the north–south extension of the Earth's crust, with extension rates estimated at around 10–14 mm a^{-1} (Ambraseys, 1997; Briole et al., 2000; Clarke et al., 1998; Floyd et al., 2010; Rigo et al., 1996).

In addition to the Corinth Rift, other extensional rifts include the Gulf of Evia and the Sperchios Basin. These Central Greece Rift Systems are part of the broader Aegean extensional regime, with active normal faults that contribute to the seismicity and crustal stretching in the broader area.

Neotectonic research and the analysis of focal mechanisms from shallow earthquakes suggest that active faults in Greece can be classified into the following types: (a) low-angle thrust faults and steeper reverse faults along the Hellenic trench; (b) east–west oriented normal faults across the Greek peninsula, the back-arc Aegean region, and western Türkiye; (c) north–south trending normal faults along the Hellenic orogen and the upper Aegean plate, primarily south of the Hellenic volcanic arc; and (d) strike-slip faults in the Northern Aegean, Epirus, the Ionian Islands, and western Peloponnese (Pavlides et al., 2024).

Greece is divided into several NW-SE trending geotectonic zones, reflecting its Alpine orogenic history (Papanikolaou, 2021b). The Internal Zones include the Pelagonian Zone, the metamorphic-rich Attic-Cycladic Massif, and the high-grade Rhodope Massif, representing the core of the Hellenides (Papanikolaou, 2021a). The External Zones consist of the Pindos Zone, with deep-water Tethyan sediments, the Ionian Zone, known for its thick carbonate platform, and the Gavrovo-Tripolis Zone, which transitions between them (Papanikolaou, 2013). Intermediate units like the Sub-Pelagonian Zone contain ophiolites and volcanic remnants of the Tethyan oceanic crust. Ophiolites are prominent across various zones, evidencing Jurassic obduction events (Papanikolaou, 2021a).

2.3 Climatic setting

Greece's climate is predominantly Mediterranean, characterized by hot, dry summers and mild, wet winters. This climate is influenced by the country's varied topography, which includes mountainous regions, coastal areas, and numerous islands. According to the Hellenic National Meteorological Service (HNMS), mean annual temperatures vary significantly across the country, averaging around 10°C in the northern regions and reaching up to 20°C in the southern islands (Figure 3a). Precipitation patterns are also variable; the western regions, such as Epirus, the Western Peloponnese, and the Ionian Islands, receive considerable rainfall – up to 2,000 mm annually – while the eastern areas, including the Aegean islands, may see as little as 400 mm (Figure 3b).

3. Data and methods

The geomorphological map was developed using a combination of digital spatial datasets, including

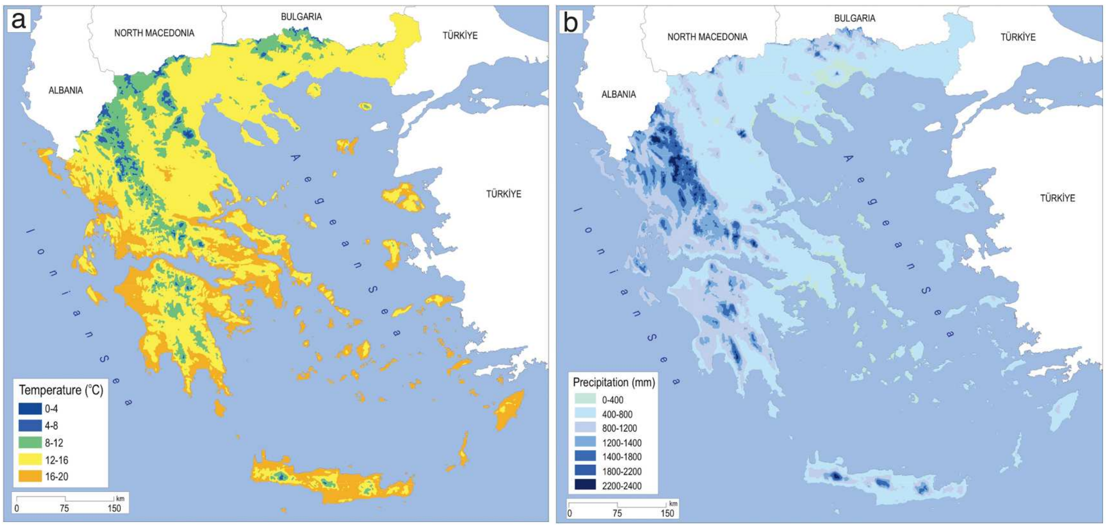


Figure 3. a. Average annual temperature of Greece over the time period 1971–2021. b. Average annual precipitation of Greece, over the time period 1971–2021 (Hellenic National Meteorological Service).

Table 1. Summary of digital spatial datasets used for the geomorphological map, including data sources, descriptions, and resolutions.

Data type	Description	Source	Resolution/ Scale/Version
Topographic	Digital Elevation Model (DEM)	Hellenic Cadastre	5 m (original), resampled to 25 m
Geological	Geological sheets of Greece	Hellenic Survey of Geology and Mineral Exploration	1:50,000
Climatic	Raster data for precipitation and temperature (1971-2021)	Hellenic National Meteorological Service	~730 m
Bathymetric	Bathymetric data for marine areas	Hellenic Centre for Marine Research	100 m
Tectonic	Vector data representing faults	National Observatory of Athens	V. 5.0

topographic, geological, tectonic, climatic, and bathymetric data. Topographic data included a 5 m resolution Digital Elevation Model (DEM), resampled to 25 m. Geological data were derived from the 1:50,000 scale geological sheets of Greece (116 sheets) provided by the Hellenic Survey of Geology and Mineral Exploration (HSGME). The geological formations of the country were categorized into six main groups: Quaternary formations, consisting of relatively recent sedimentary deposits, cover 22.6% of the total area. Neogene formations account for 15%, while carbonate rocks, which are integral to the development of the country's karst landscapes, represent 23.6%. Igneous rocks make up 15.5%, and metamorphic rocks contribute 12.3%. Lastly, flysch and molasse formations cover 11% of the total area. Climatic data, consisting of raster data for precipitation and temperature, were obtained from the Hellenic National Meteorological Service for the period 1971-2021. Bathymetric data were sourced from the Hellenic Centre for Marine Research. Additionally, faults were incorporated from vector datasets made available by the National Observatory of Athens (Ganas et al., 2013). Details of the datasets are summarized in Table 1.

The identification and delineation of the landforms were based on the literature review of previous geomorphological studies at various scales, as well as on the application of semi-automated, GIS techniques. The methodological procedures can be divided into three phases (Figure 4).

1st phase: *Collection and preparation of literature and data, determination of the appropriate methodology and mapping protocols, and organization of available data into a GIS spatial geodatabase.* High resolution topography obtained from the Hellenic Cadastre, as well as the geological maps of Greece on a scale of 1:50,000, obtained from the Hellenic Survey of Geology and Mineral Exploration, were used during this phase as initial inputs in the geodatabase for the production of a series of derivatives, which included a hillshade map, a slope-aspect map, and a Red Relief Image Map (RRIM).

2nd phase: *Identification and delineation of morphological features based on the GIS analyses performed in the previous phase.* Landforms such as alluvial fans and cones, gravity-induced forms, planation surfaces, breaks of slope, karst depressions etc., were identified. The preliminary morphological sketch produced during this phase was evaluated using Google Earth imagery for the interpretation of the morphological features and the determination of the landforms.

3rd phase: *Comprehensive evaluation of the morphological sketch.* The graphic design process included the clear and systematic symbolization of geological formations and landforms. Geomorphological symbols suggested by Campobasso et al. (2018) were applied ensuring adherence to international standards. Specific colors were assigned to each category of landforms allowing for easy recognition of the dominant geomorphological processes in different parts of the

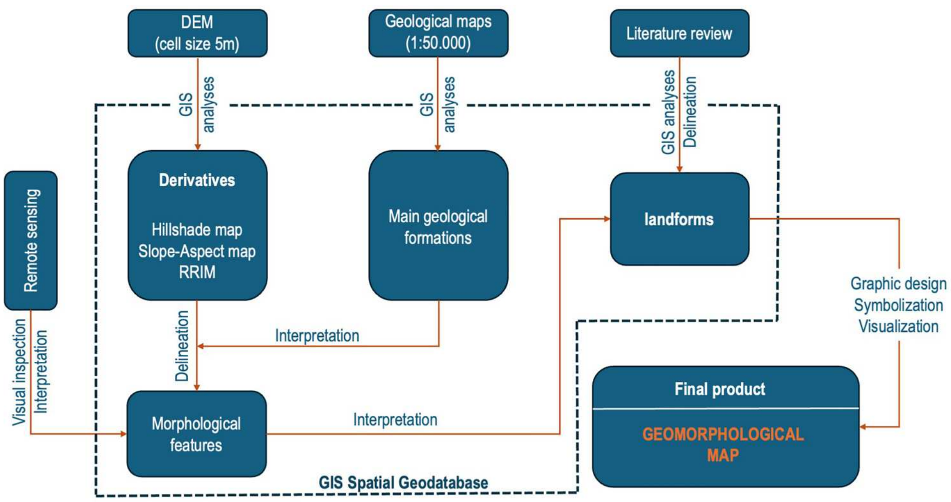
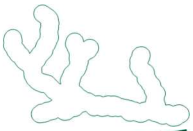
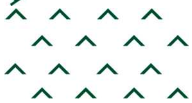
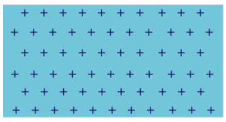
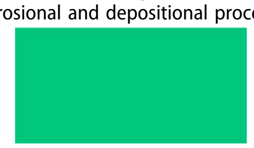
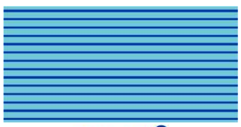
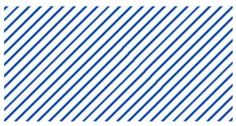


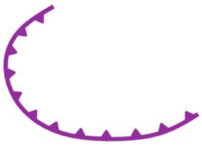
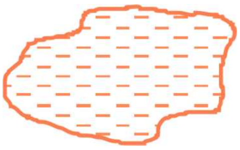
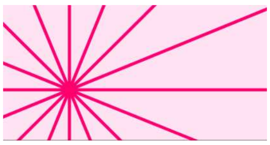
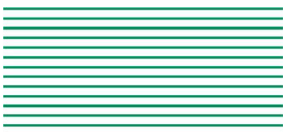
Figure 4. Flow diagram outlying the main methodological steps of the map production.

Table 2. Summary of mapped geomorphological features with symbols, data sources, and spatial representation.

No.	Landform/ Feature	Symbol	Feature type	Data source	References/Techniques
A. Structural landforms					
A_1	Ridge crest		Line	GIS Analysis	Red Relief Image Map
B. Fluvial erosion landforms					
B_1	Intense incision		Line	GIS Analysis	Red Relief Image Map
B_2	Gorge		Line	GIS Analysis	Red Relief Image Map
B_3	Badland		Polygon	GIS Analysis	Red Relief Image Map
C. Fluvial deposition landforms					
C_1	River delta		Polygon	Literature review	(Chabrol et al., 2022; Karditsa et al., 2020; Karymbalis et al., 2016; Karymbalis et al., 2018; Karymbalis, Gallousi, Cundy, Tsanakas, et al., 2022; Maroukian & Karymbalis, 2004; Parcharidis et al., 2011; Poulos et al., 2022)
C_2	Alluvial fan		Polygon	Literature review	(Karymbalis et al., 2018, 2022a, 2022b, 2022c, 2022d; Karymbalis et al., 2018)
C_3	Fan delta		Polygon	Literature review	(Karymbalis et al., 2018) Hellenic Survey of Geology and Mineral Exploration Geological sheets of Greece (1:50.000)
D. Landforms due to fluvial erosional and depositional processes					
D_1	Fluvial terraces		Polygon	Literature review	Hellenic Survey of Geology and Mineral Exploration Geological sheets of Greece (1:50.000)
E. Coastal landforms					
E_1	Coastal plain		Polygon	GIS Analysis	(Ghilardi et al., 2008; Karymbalis, Tsanakas, Cundy, Iliopoulos, et al., 2022; Tsanakas et al., 2019a, 2019b)
E_2	Marine notches	0.8-(+1.6) ○	Point	Literature review	(Gaki-Papanastassiou et al., 2007; H. Maroukian et al., 2000; Maroukian et al., 2008; P.A. Pirazzoli et al., 1989, 1999; Pirazzoli et al., 1994; S. Papageorgiou et al., 1993; Stiros et al., 2000; Stiros & Pirazzoli, 1995)
E_3	Marine terraces		Polygon	Literature review	(Athanassas & Fountoulis, 2013; De Gelder et al., 2015, 2022; Gaki-Papanastassiou et al., 2011, 2007; Gallen et al., 2014; Karymbalis, Tsanakas, Cundy, Iliopoulos, et al., 2022; Karymbalis, Tsanakas, Tsodoulos, Gaki-Papanastassiou, et al., 2022; Ott et al., 2019; Papanastassiou et al., 2014; Tsanakas et al., 2022)
E_4	Coastal slope	 <10° 10° – 30°  >30°	Polygon	GIS Analysis	Slope-Based Coastal Morphology Classification using TIN Analysis

(*Continued*)

Table 2. Continued.

No.	Landform/ Feature	Symbol	Feature type	Data source	References/Techniques
F. Glacial and periglacial landforms					
F_1	Cirque		Line	Literature review	(Bathrellos et al., 2014; Nance, 2010; Pavlopoulos et al., 2018; Pope et al., 2017; Smith et al., 1997; Styllas et al., 2018; Woodward et al., 2004)
F_2	Glacial deposits		Polygon	Literature review	(Bathrellos et al., 2014; Nance, 2010; Pavlopoulos et al., 2018; Pope et al., 2017; Smith et al., 1997; Styllas et al., 2018; Woodward et al., 2004)
F_3	Glaciofluvial deposits		Polygon	Literature review	(Nance, 2010; Pavlopoulos et al., 2018; Pope et al., 2017; Smith et al., 1997; Styllas et al., 2018; Woodward et al., 2004)
G. Karst landforms					
G_1	Polje		Polygon	GIS analysis	Red Relief Image Map
G_2	Cave		Point	Literature review	(Alexakis et al., 2021; Bathrellos et al., 2024; I.K. Giannopoulos et al., 2023; Konsolaki et al., 2020; Pennos et al., 2014, 2019; Vassilakis & Konsolaki, 2022)
H. Gravitational landforms					
H_1	Landslide		Point	Literature review	(Bathrellos et al., 2009; Bliona, 2008)
I. Aeolian deposition landforms					
I_1	Dune field		Polygon	Literature review	Hellenic Survey of Geology and Mineral Exploration Geological sheets of Greece (1:50.000)
J. Volcanic landforms					
J_1	Volcano		Polygon	Literature review	(Dietrich & Lagios, 2018; D'Alessandro et al., 2009; Parks et al., 2012; Tzanis et al., 2020; Zhou et al., 2021)
K. Other landforms					
K_1	Planation surface		Polygon	GIS Analysis	Identification and Classification of Planation Surfaces via Morphological Slope Mapping
K_2	Depositional surface		Polygon	GIS Analysis	Identification and Classification of Depositional Surfaces via Morphological Slope Mapping
K_3	Knickpoint		Point	GIS Analysis	Knickpoint Identification and Mapping through RDE Index Analysis
K_4	Capture elbow		Point	Literature review	(Zelilidis, 2000)
K_5	Windgap/ Drainage reverse	 	Point/ Polygon	Literature review	(Zelilidis, 2000)

landscape (Table 2). To prevent visual confusion with the geomorphological features, light colors were used for the geological formations.

3.1 Red Relief Image Map (RRIM)

This method is based on multi-layered topographic information computed from three – dimensional data (DEM). The basic concept is the multiplication of three element layers, topographic slope, positive, and negative openness (Chiba et al., 2008). Positive and negative openness were defined by (Yokoyama et al., 2002). Negative openness represents concavities of surfaces and takes higher values on landforms such as valleys, gorges, or the inside parts of craters (Figure 5a). On the other hand, positive openness represents convexities and is ideal for mapping crests, ridges, scarps etc (Figure 5b). In this way, the Red Relief Image Map can visualize not only topographic slopes, but also concavities and convexities at the same time.

The generation of the RRIM is based primarily on the calculation of the Ridge and Valley Index (RVi) (Chiba et al., 2008) as shown below:

$$RVi = \frac{Op - On}{2}$$

where O_p is the positive openness and O_n is the negative openness as defined by Yokoyama et al. (2002). The RVi overcomes the dependency on incident light direction, unlike shaded relief images, while capturing both convexity and concavity. It assigns high values to convex features and low values to concave ones. In RRIM, this information is displayed using a grayscale layer for topographic expression and a red layer for slope (Figure 5c). By effectively depicting landforms at multiple scales, RRIM has been utilized in this study to map geomorphological features such as ridge crests, intense incision, flat surfaces, gorges, and badlands.

3.2 Identification and mapping of knickpoints

To identify and map knickpoints within the study area, the Relation Declivity Extension (RDE) index was considered. The RDE index was developed by (Etchebehere et al., 2004) as a derivative of the Hack Index and reflects the energy present in each drainage stretch based on surface slope and water discharge. This index is calculated by dividing the RDE index for a specific stretch (RDEs) by the total RDE index for the entire drainage area (RDEt) (Figure 6) as indicated below:

$$RDE = \frac{RDEs}{RDEt}$$

$$RDEs = \left(\frac{\Delta h}{\Delta l}\right) * L$$

$$RDEt = \left(\frac{\Delta h}{\Delta l}\right) * \ln(L)$$

For the identification of the knickpoints the focus was placed on stretches with high (RDE) index values, particularly those exhibiting RDE anomalies greater than 10.

The RDE indices of various channel stretches and their respective drainage basins were measured

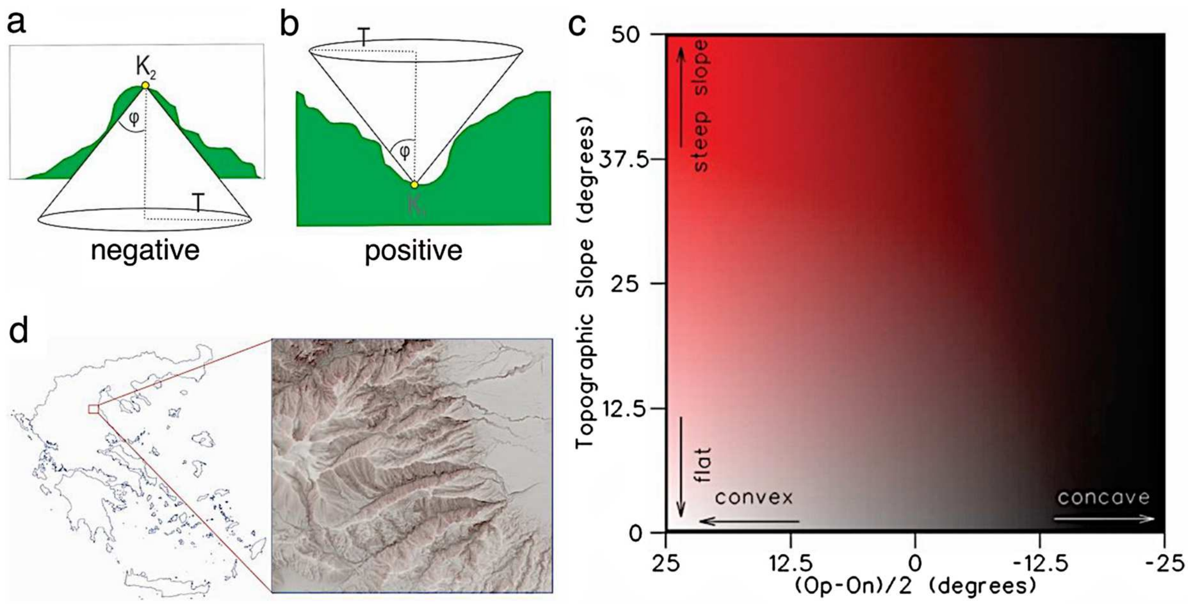


Figure 5. Conceptual diagram of negative openness (a) and positive openness (b), modified by (Yokoyama et al., 2002). c) Color diagram of RRIM. Topographic slope is showed as chroma value of red (y axis) and (Op-On)/2 is showed as brightness (x axis) (Chiba et al., 2008). d) Red Relief Image Map (RRIM) of the eastern flank of Mount Olympus. The visualization emphasizes topographic details such as ridges, valleys, and escarpments.

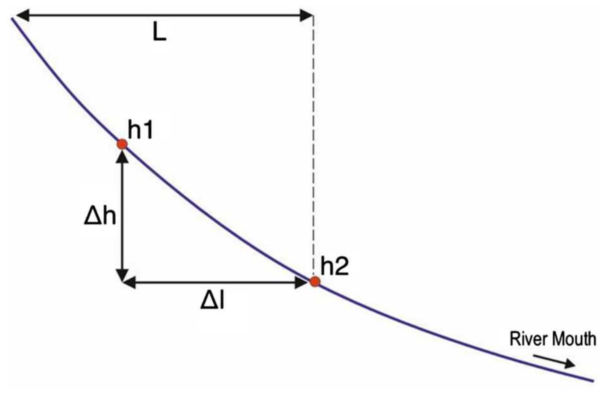


Figure 6. Diagram illustrating key parameters used in the knickpoint identification methodology.

utilizing the Knickpoint Finder software (Queiroz et al., 2015). This software enabled the comparison of RDEs/RDEt values to identify anomalous slopes indicative of knickpoints. The analysis adhered to established thresholds: stretches with an RDEs/RDEt ratio greater than 2 were classified as anomalous, with second-order anomalies falling between 2 and 10, and first-order anomalies identified where RDEs/RDEt exceeded 10. This threshold-based approach highlights areas with abrupt changes in channel slope, enabling the systematic detection and mapping of major knickpoints.

3.3 Identification and mapping of planation and depositional surfaces

The first step involved defining typical horizontal surfaces, which are extensive and easily identifiable areas with an average slope lower than that of the surrounding regions. The morphological slope map and slope direction map served as fundamental analysis maps, either independently or in combination with geological maps, for locating and identifying morphological features such as horizontal or near-horizontal flattening surfaces, sloping surfaces, and morphological discontinuities often associated with tectonic structures. The mapping process, described below, is based on the methodology developed by Vassilakis (2006).

The identified surfaces are characterized by slopes ranging from 0° to 1° and are typically grouped according to absolute altitude. They are classified as depositional surfaces when linked to sediment deposition (post-Alpine formations) and as planation surfaces when formed by the erosion of Alpine formations. To identify and map these surfaces, the morphological slope map was utilized, where desired slopes were selected (0° to 1°), and the mosaic data were converted into vector data. Additional characteristics of each surface were calculated, including area, perimeter, minimum, maximum, and average altitude, as well as the range (the difference between maximum and minimum values). Finally, only the surfaces deemed reliable and representative for the current analysis were selected – specifically, those with an area greater than 0.01 km².

3.4 Mapping coastal morphology

To identify coastal morphology, a Triangulated Irregular Network (TIN) surface was developed using contour lines with a 4 m interval. The triangle faces from the TIN dataset were exported to obtain slope attributes for each triangle, focusing on a buffer area between the coastline and the 20 m contour line. These slope values were then assigned to the coastline, which was ultimately categorized into three groups: less than 10°, between 10° and 30°, and greater than 30°.

4. Results

The Main Map reveals a complex spatial distribution of landforms and geomorphological features across Greece, reflecting the interplay of endogenic and exogenic processes that shape the relief. Ridge crests are predominantly found in mountainous regions (i.e. Pindos Mountain range, central Peloponnese and other) characterized by pronounced relief and are closely associated with the underlying geological structure of the basement rocks (Figure 7b).

Intense incision is a notable fluvial geomorphic process observed in Greece, particularly in regions characterized by intense tectonic uplift. In many cases it is associated with the formation of deep gorges, which develop in areas where the lithology is resistant to erosion (Figure 7g, Figure 8a, d). The cases of Vikos Gorge in Epirus, Vouraikos gorge in north Peloponnese, and Samaria Gorge in Crete among others exemplify this relationship, demonstrating how intense tectonic uplift and subsequent river downcutting can create deep valleys with very steep slopes.

Badlands, predominantly located in the northern Peloponnese, are formed in areas with easily erodible lithologies like marls (Figure 9b). Their formation is the result of rapid weathering and erosion processes, intensified by the region's climatic conditions. River deltas, alluvial fans and fan deltas are found at the mouths of perennial rivers (i.e. Aliakmonas, Acheloos, Pineios, Evros etc.) and intermittent torrents where sediment transported by flowing water is deposited upon reaching a standing body of water or a gentler slope (Figure 9d). They are important for shaping coastal environments and are influenced by factors such as sediment load, river discharge, oceanographic conditions (waves, currents, tides) and the bathymetry of the receiving basins (Chabrol et al., 2022; Karditsa et al., 2020; Karymbalis et al., 2016; Karymbalis, et al., 2022a, 2022b, 2022c, 2022d; Karymbalis et al., 2018a, 2018b; Maroukian & Karymbalis, 2004; Parcharidis et al., 2011; Poulos et al., 2022).

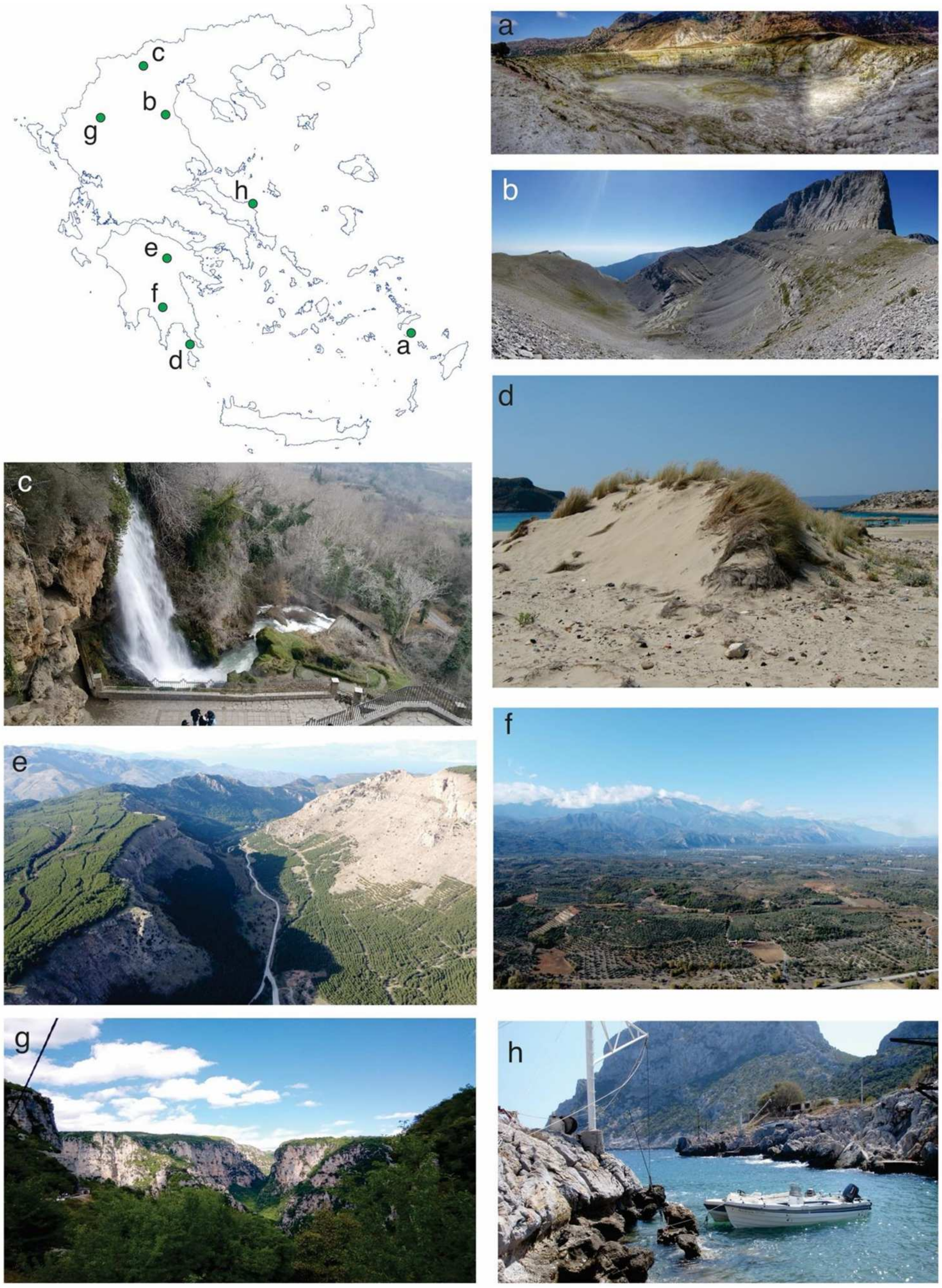


Figure 7. a) Panoramic view of a crater on Nissyros volcanic island, b) Glacial cirque at the summit of Mount Olympus, c) Knick-point along the Edesseos River (Edessa waterfalls), d) Sand dunes on Elafonissos Island, e) Wind gap in northern Peloponnese associated with the Dervenios and Olvios drainage reversal, f) Taygetos mountain front and Evrotas river alluvial plain, g) Vikos Gorge in northwestern Greece, h) Uplifted tidal notch in Chili (eastern Evia Island).

Fluvial terraces are an important feature found in river valleys across several parts of the country. They were mapped in locations where the scale of the geomorphological mapping permitted their identification. They are formed as a result from repeated cycles of river deposition and incision and are associated with past climatic fluctuations and/or tectonic uplift.

Coastal plains in Greece, are mainly characterized by gentle slopes of less than 10 degrees. They are typically associated with low-lying flat areas consisting of unconsolidated sediments deposited by fluvial systems in coastal areas (Ghilardi et al., 2008; Karymbalis, Tsanakas, Karkani, & Evelpidou, 2022; Tsanakas et al., 2019a, 2019b).

Tidal notches have been mapped along cliffed coasts composed of hard rocks (Figure 7h), primarily limestones, in tectonically active areas such as the Ionian islands (Lefkada, Cephalonia, Corfu and Zakynthos), as well as along the coastlines of the Gulf of Corinth, North Evoikos Gulf, and various locations around the Islands of Crete, Rhodes, and Samos. They are found both uplifted and submerged, indicating co-seismic vertical tectonic movements during the late Holocene and reflecting the area's tectonic and seismic history (Gaki-Papanastassiou et al., 2007; Maroukian et al., 2000; Maroukian et al., 2008; Papageorgiou et al., 1993; Pirazzoli et al., 1989, 1999; Pirazzoli et al., 1994; Stiros et al., 2000; Stiros & Pirazzoli, 1995).

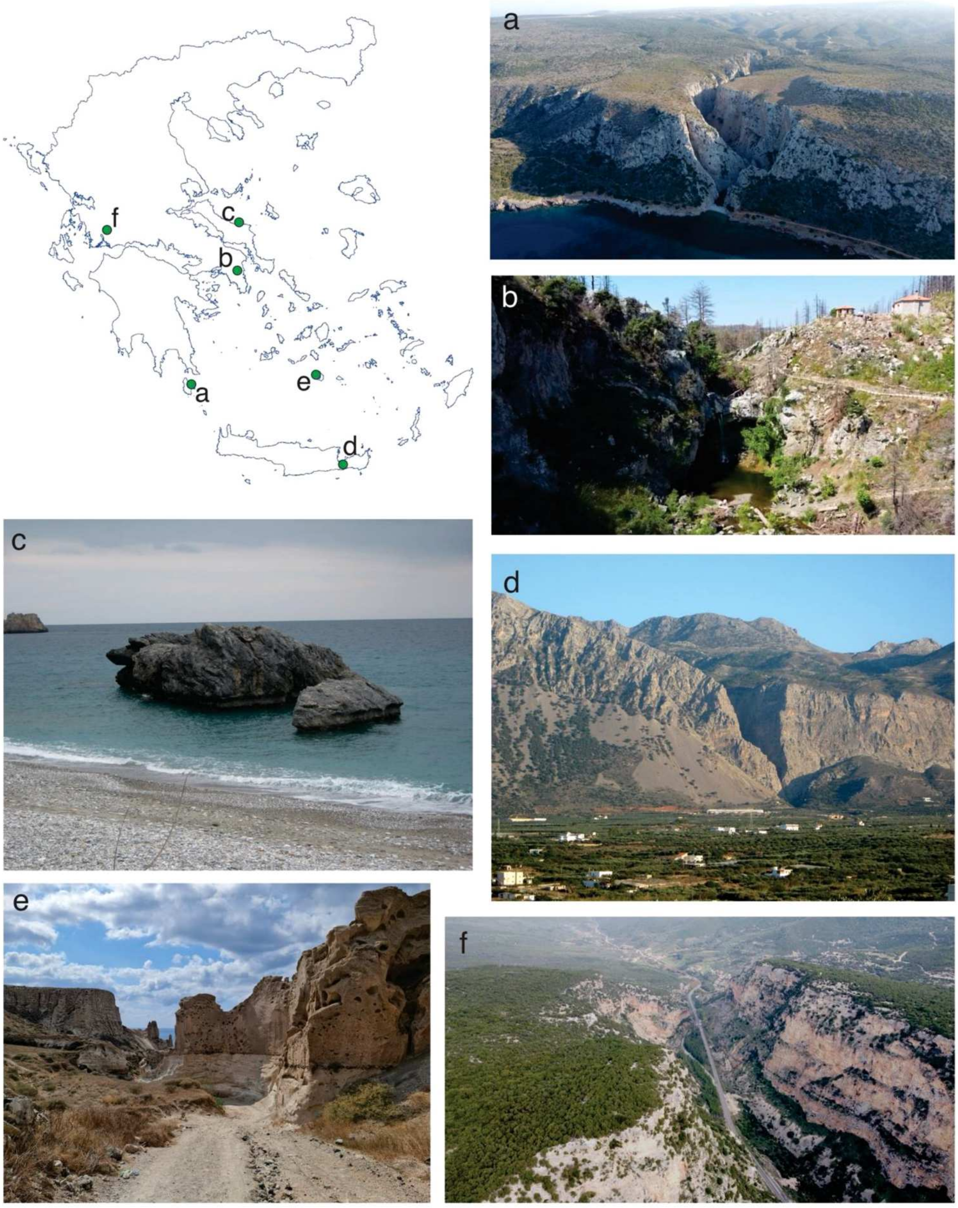


Figure 8. a) Kakia Lagada gorge located in eastern Kithira Isl. b) Knickpoint at a Megalo Rema River tributary in eastern Attica, c) Uplifted notch at Hiliadou (eastern Evia Island), d) Ha River Gorge in northeastern Crete Island, e) Pumice formations on the volcanic island of Santorini, f) Kleissoura wind gap in central Greece.

Uplifted marine terraces, which mark levels of palaeoshorelines corresponding to past sea-level highstands, indicate tectonic uplift during the Late Pleistocene and are present in locations mainly along the Hellenic Subduction Zone, including Cephalonia Island, the Messiniakos and Lakonikos Gulfs, the islands of Kithira, Crete, Karpathos and Rhodes, as well as other tectonically active areas like the Gulf of Corinth and North Evoikos Gulf (Figure 9e,g) (Athanassas & Fountoulis, 2013; De Gelder et al., 2015, 2022; Gaki-Papanastassiou et al., 2011; Gallen et al., 2014; Karymbalis, Tsanakas, Karkani, & Evelpidou, 2022; Karymbalis, Tsanakas, Karkani, & Evelpidou, 2022; Ott et al., 2019; Papanastassiou et al., 2014; Tsanakas et al., 2022a, 2022b). They have formed through the combined effects of wave erosion, sedimentation, and tectonic uplift and late Pleistocene sea-level fluctuations.

The coasts of the Greek territory were categorized into three morphological slope categories: (i) coasts with slopes less than 10 degrees, which are often associated with depositional features like coastal plains, deltas, and fan deltas and account for 52.6% of the total coastline length; (ii) coasts with slopes ranging between 10 and 30 degrees, representing 27.9% of the coastline; and (iii) coasts with slopes greater than 30 degrees, typically linked to coastal cliffs and rugged coastal terrains, covering 19.5% of the total coastline length.

In Greece, glacial landforms are relatively scarce, with the most prominent examples found in high-altitude regions such as Mount Olympus (Figure 7b), the Pindos Mountain range, and Mount Helmos. The

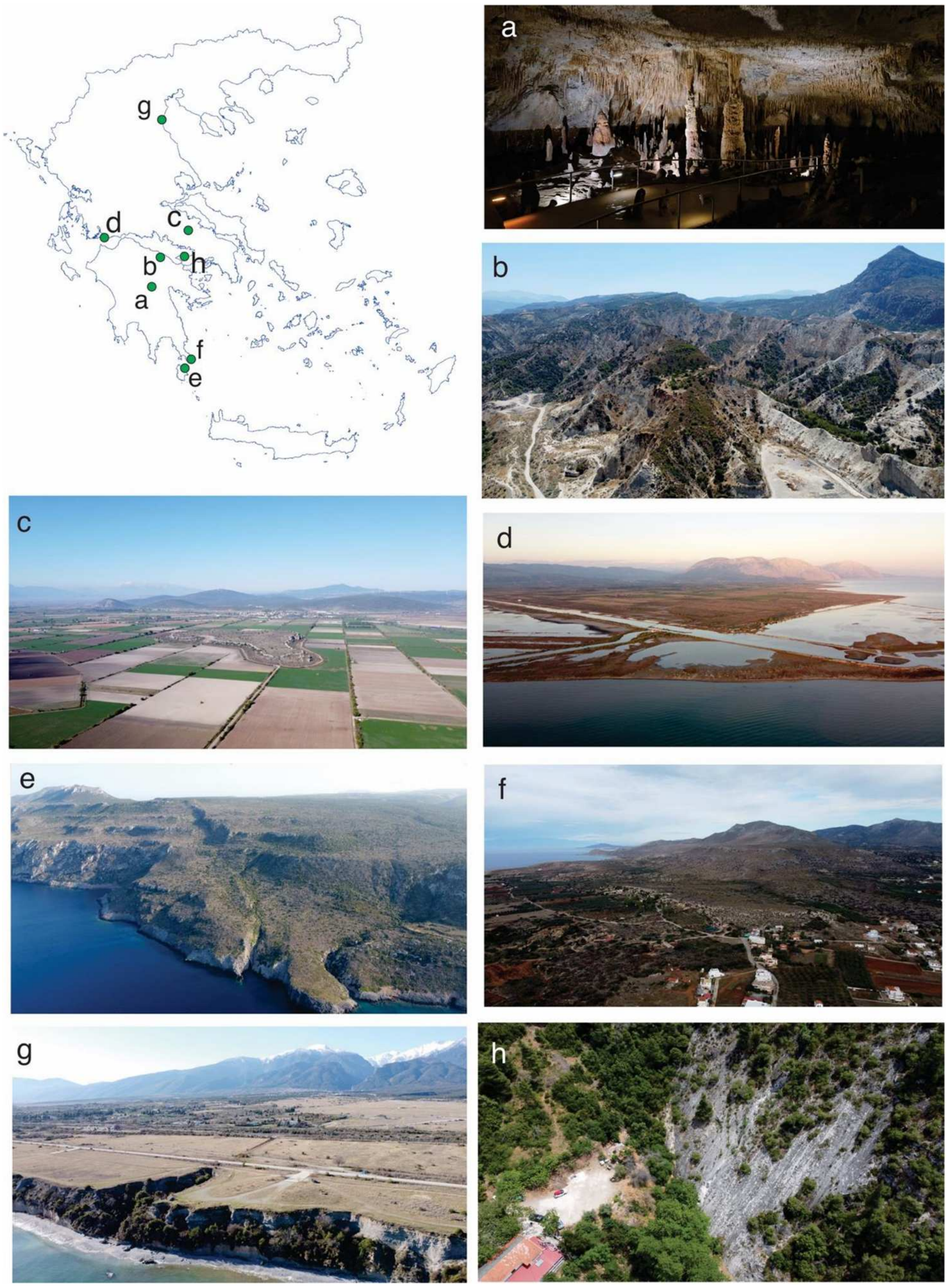


Figure 9. a) Kapsia Cave in central Peloponnese, b) Badlands in northern Peloponnese, c) Kopaida polje in central Greece, d) Evinos River delta at the northern Patraikos Gulf, e) Uplifted marine terraces on eastern Kithira Island deformed by an active fault, f) Uplifted marine terraces at Cape Maleas, g) Alluvial fans at the foot of Mount Olympus in northern Greece, h) Pissia active normal fault.

presence of these features indicate a history of glaciation during late Pleistocene glacial periods that has sculpted the landscape in relatively restricted places of the country (Bathrellos et al., 2014; Nance, 2010; Pavlopoulos et al., 2018; Pope et al., 2017; Smith et al., 1997; Styllas et al., 2018; Woodward et al., 2004).

Karst landforms, primarily developed on carbonate rocks, are widespread in northern and central Greece (Figure 9a,c), the Peloponnese, and numerous islands (Alexakis et al., 2021; Bathrellos et al., 2024; Giannopoulos et al., 2023; Konsolaki et al., 2020; Pennos et al., 2014, 2019; Vassilakis & Konsolaki, 2022). The region's climatic conditions – characterized by seasonal precipitation and temperature variations – have fostered the development of these features through processes of chemical weathering and dissolution, leading to distinctive landforms such as poljes, dolines, sinkholes, caves, and underground drainage systems.

Landslides are prevalent in mainland Greece and the Peloponnese, particularly along the Pindos Mountain range, where a combination of geological conditions (such as alternations of rock layers with diverse lithology) and tectonic activity along with high precipitation create environments prone to slope failure (Bathrellos et al., 2009; Bliona, 2008). Earthquakes, a frequent occurrence in these tectonically active regions, serve as significant triggers for landslides, especially in areas where the lithology is prone to instability.

Coastal sand dune fields, such as those mapped along the coastline of western Peloponnese (notably at Kaiafas and Elafonissos Isl.), are formed in areas with abundant sand supply, consistent wind patterns, and vegetation that helps stabilize the dunes (Figure 7d).

Active volcanoes are found along the volcanic arc of Greece, including those at Sousaki, Methana, Milos, Santorini, and Nissyros (Dietrich & Lagios, 2018; D'Alessandro et al., 2009; Parks et al., 2012; Tzanis et al., 2020; Zhou et al., 2021) (Figure 7a, Figure 8e). They are associated with the subduction zone where the African plate is converging with the Eurasian plate. The most active ones are those of Santorini and Nisyros.

Planation surfaces were mapped in several areas of western and central Macedonia, Peloponnese highlands, in Thrace and Crete. They are linked with erosional processes during long periods of tectonic stability that allows extensive flattening of the landscape. In many cases, planation surfaces are linked to karst processes and host various karst depressions such as dolines and uvalas. Depositional surfaces in Greece are mainly associated with depositional processes in various environments like fluvial, lacustrine, and karst.

Major knickpoints, often associated with significant waterfalls, were identified along rivers and typically result from normal faulting or differential erosion that creates abrupt changes in river gradient (Figure 7c, Figure 8b).

Capture elbows and wind gaps are prominent features in the northern Peloponnese and Evia Island, both attributed to rapid and intense tectonic activity (Zelilidis, 2000). Wind gaps were mapped in northern Peloponnese and central Greece (Zelilidis, 2000) (Figure 7e, Figure 8f).

5. Discussion and conclusions

The aim of this paper was to produce a geomorphological map of Greece at a scale of 1:1,000,000. The Main Map is an attempt to identify and depict the major landforms across the country, classifying them according to the dominant processes that contributed to their formation and evolution. This national-scale landforms classification is an important step towards a comprehensive geomorphological inventory. The resulting Main Map gives information on the long-term geomorphological history of the region, revealing that the landscape is characterized by diverse geomorphic environments, configured primarily by the combined action of active tectonics and exogenic processes.

The methodological approach has shown that small-scale geomorphological mapping faces challenges in balancing detail with computational efficiency, necessitating more advanced and effective GIS techniques. Future projects should prioritize optimizing GIS tools to manage increasingly large datasets effectively. Additionally, innovative symbolization approaches are essential to address the challenges of generalizing landforms originally mapped at larger scales.

The Main Map can serve as a valuable reference for interdisciplinary research on geological risks assessment, risk mitigation, including those related to climate change – and sustainable development at the national scale. It can also be a practical management tool to support land-use planning, environmental conservation efforts, and infrastructure development.

To ensure the continuous updating of the geomorphological information provided by this small-scale Main Map, the development of a Web GIS platform represents a promising solution and is one of our primary future goals. Such a platform could address the challenge of visualizing large-scale landforms at higher zoom levels, maintaining detailed representation across multiple scales. Additionally, incorporating geomorphological information into a web-GIS platform could enable detailed landforms to be displayed in a way that is both accurate (truthful to the landscape) and readable (easy to interpret) even when viewed at smaller scales. This approach would also support data sharing and collaboration among researchers, policymakers, and the public, fostering a more accessible and integrated understanding of geomorphological features.

Software

The production of the geomorphological map utilized ArcGIS 5.1 for spatial analysis and cartographic workflows, QGIS 3.28 for additional GIS processing and data integration, and CorelDRAW Graphics Suite 2023 for final map design, symbolization and layout adjustments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request. Requests for data will be assessed on a case-by-case basis to ensure compliance with ethical, privacy, and security considerations.

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